DAEN 427 Decision and Risk Analysis

Fall 2026

Course Information

Time (Location) Mon-Wed-Fri 11:30–12:20 AM *3 credit hours* (ZACH 344)
Prerequisites Grade of C or better in ISEN 310, DAEN 321, or STAT 212
Cross listing ISEN 427 / ISEN 627
Instructor Alexander Abuabara mailto:abuabara@tamu.edu
Course office hours (location) Wed 1-3 PM or by appointment (ETB 4013)

Description

Overview of the state of the art in descriptive and prescriptive theories of decision making under uncertainty with emphasis on the ways in which human decisions depart from normative models of rationality; analytical foundations stemming from several disciplines, economics, psychology, management science; application in engineering systems will be considered.

Outcomes

1. Apply Bayesian approach to perception and decisions under uncertainty.
2. Implement computational models for Bayesian inference and strategic decision support.
3. Use concepts from economics, psychology, and management science to analyze decision-making processes.

Recommended Resources

- Bayes Rules! An Introduction to Applied Bayesian Modeling https://www.bayesrulesbook.com/.
- ... Bayesian Inference, Methods, Computation https://link.springer.com/book/10.1007/978-3-030-82808-0.
- A Course in Behavioral Economics https://www.erikangner.com/a-course-in-behavioral-economics.
- R https://cran.r-project.org/, RStudio (IDE) https://posit.co/download/rstudio-desktop.

Grading

Assessment	**Weight**	**Description**
Exam (2)	58% (29% each)	Content on weeks 1–5 & 1–11
Project (report + presentation)	28% (groups of 2)	Case study exploring decision and risk analysis in engineering
Participation / Activities	14%	In-class quizzes and discussions, assignments, practice questions

Grades are based on total points earned: A ≥ 90%, B ≥ 80%, C ≥ 70%, D ≥ 60%, F < 60%.

Schedule (*tentative*)

Week	**Topic**	**Reading**
1	Bayesian thinking for risk and decision analysis	Ch. 1
2	Bayesian updating and evidence accumulation	Ch. 2
3	Prior modeling and expert knowledge	Ch. 3–4
4	Bayesian simulation and uncertainty propagation	Ch. 5–6
5	MCMC and computational Bayesian inference	Ch. 7
Sep 28	**Exam 1: Bayesian foundations + computation**	Ch. 1–7
6	Bayesian workflow and model validation	Ch. 8
7	Bayesian regression for risk modeling	Ch. 9
8	Bayesian prediction and model comparison	Ch. 10
9	Multiple regression and decision drivers	Ch. 11
10	Bayesian count models and rare events	Ch. 12
11	Bayesian classification and logistic risk models	Ch. 13–14
Nov 9	**Exam 2: Predictive modeling + risk analysis**	Ch. 1–14
12	Hierarchical Bayesian models	Ch. 15–16
13	Advanced multilevel and structured uncertainty models	Ch. 17–18
14	Bayesian decision analysis and probabilistic forecasting	Ch. 19
Nov 30/Dec 1	**Project on decision, risk analysis, and probabilistic programming**	Ch. 1–19

Policies

- Be nice. Be honest. Don't cheat. Adhere to all University Policies.
- Course material is cumulative and requires consistent effort. All assignments must be completed carefully and on time. Presentation quality (organization, clarity, tables, graphs, and discussion) is part of grading.
- Excused absences apply only to attendance and in-class activities and do not extend assignment deadlines. Alternative arrangements must be made prior to the deadline.
- Regrade applies to the entire submission. To ensure fairness and timely feedback, no changes after one week.

- Email is the primary form of communication; include the course number in the subject line. Responses should not be expected after 5:00 PM, on weekends, or on holidays.
- All assignments must be submitted through Canvas. If technical issues prevent submission, send your assignment from your `@tamu.edu` email to the instructor's `@tamu.edu` email as a backup. The subject line must include the course number and assignment name. Late work is not accepted and will receive a grade of zero.
- Make-up exams may be administered orally on a date set by the professor, as soon as the student is cleared to return to class following a documented absence. Documentation for excused absences must be verifiable.
- Audio or video recording of lectures or any portion of the course is prohibited without written instructor permission. Unauthorized recording constitutes a violation of student conduct.
- Headphones and non-class-related browsing are not permitted during class.
- Use of others' ideas is encouraged as part of scholarly work, but all sources must be properly cited. Proper attribution is required for all coursework and projects.